Visual processing

John Locke
Hume
Berkeley

- The understanding of visual processing has it roots in Immanuel Kant

→ "Brain's tendency to impose a pattern"

+ It begins in the Two Retinae -

Retina → Optic Nerve → Optic chiasm ⟨ temporal hemiretinas / nasal "

↓

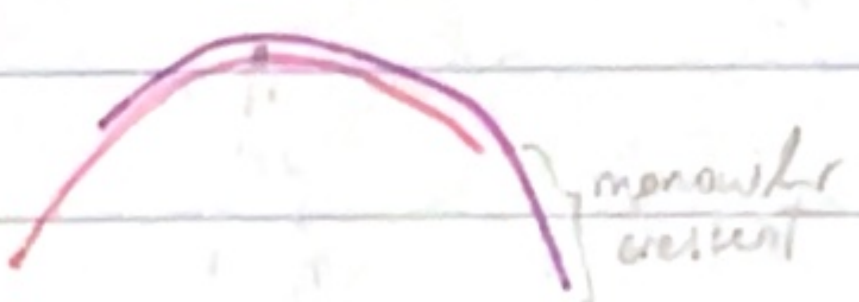
lateral geniculate nucleus of the thalamus

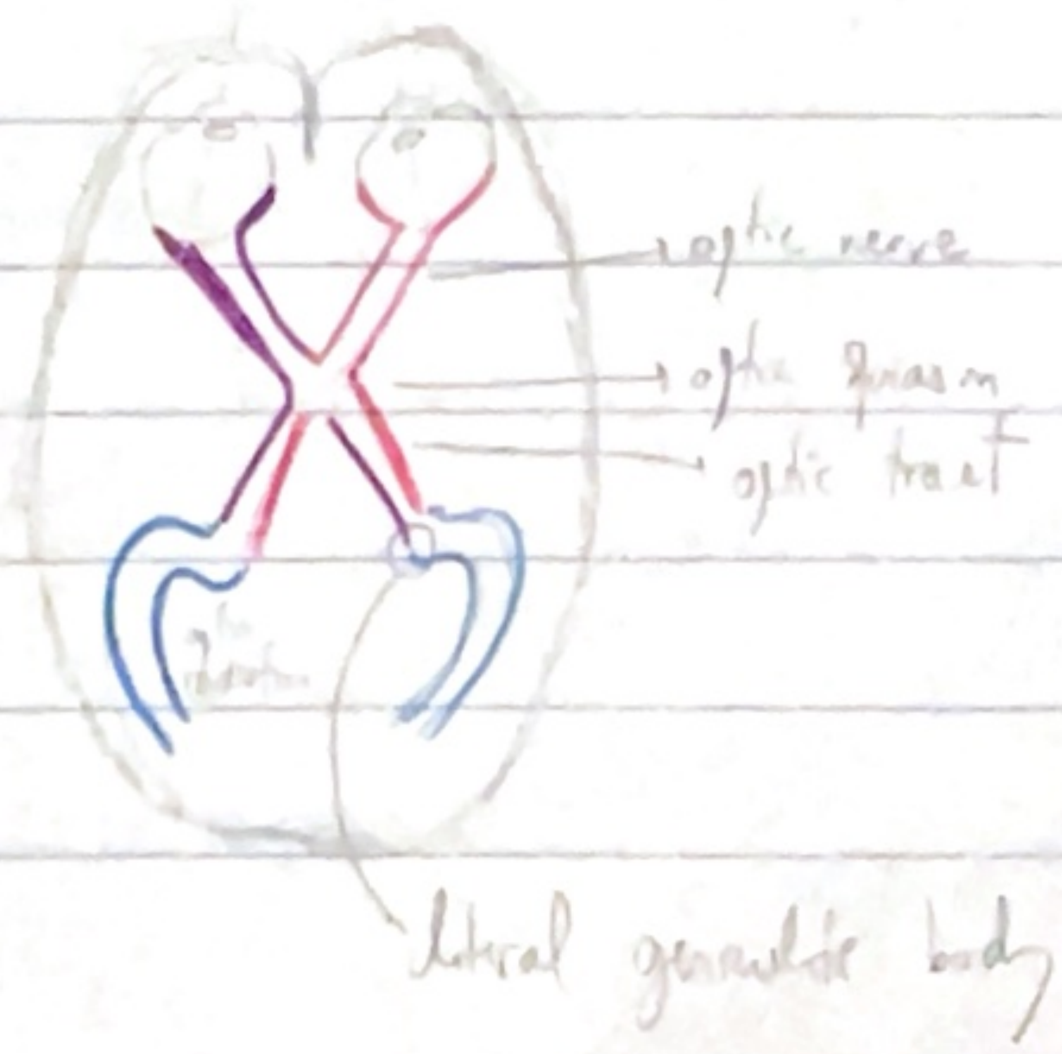
↓ optic radiations

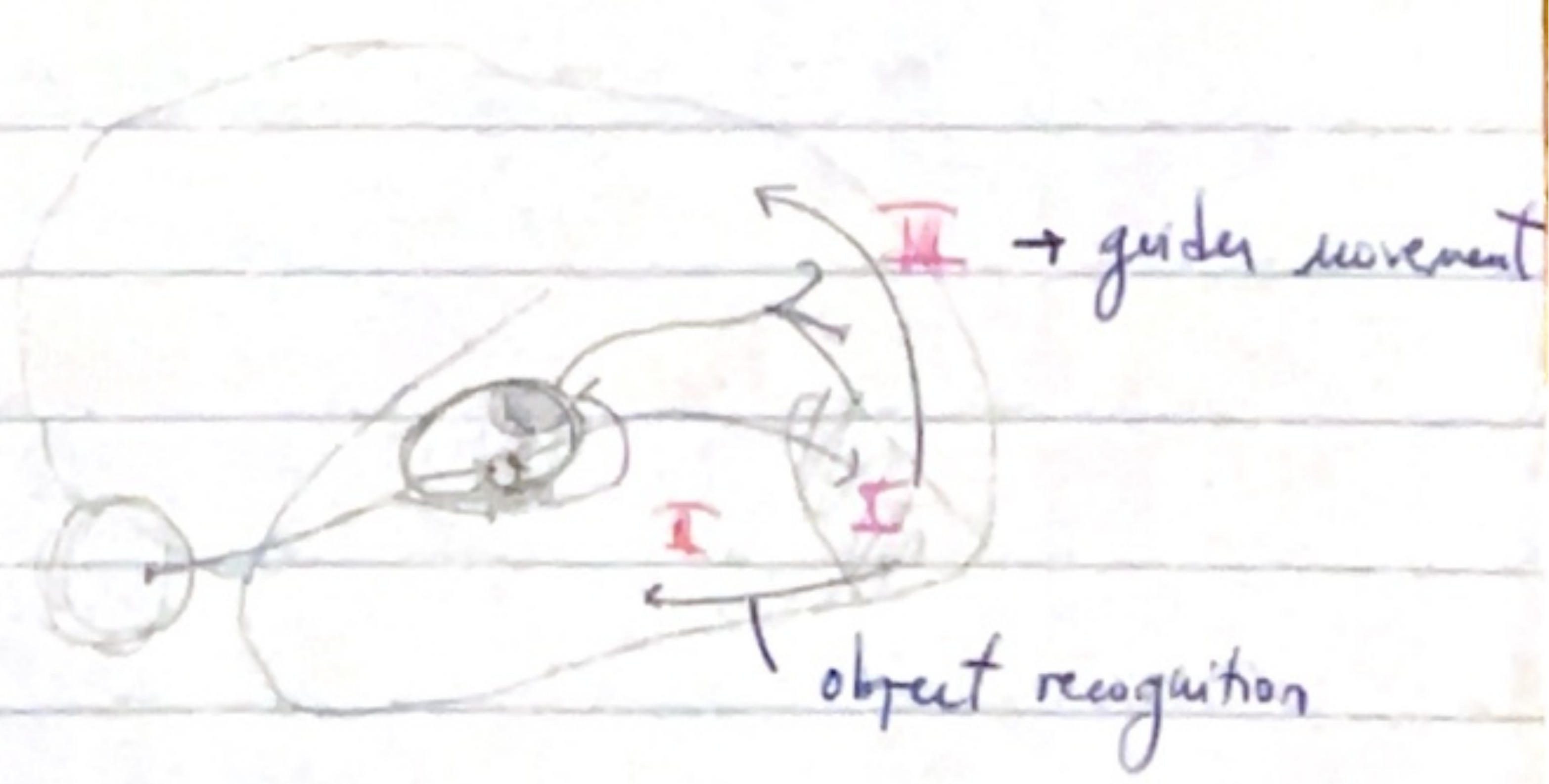
I Cortex. (Striate / 1st visual) ⟨ II temporal ventral → what / IV dorsal parietal → where

1) · Primary visual pathway → Geniculostriate pathway
   · 2nd → Superior colliculus (controls eye movement) → Reticular formation ↓ extraocular motor nuclei
   · 3rd — pretectal area of midbrain
     ↳ pupillary reflexes.

1) A visual scene is analyzed in three levels:
   1st) low level processing — orientation, colour, contrast, disparity, mov. orientation
   2) Intermediate level processing — local visual features are assembled into surfaces, objects are segregated from background, local orientation is integrated into global contours
   3) high-level processing — Identifying the object







unilateral neglect

prosopagnosia

+ Information in the dorsal pathway can contribute to object recognition through kinematic cues

↳ pathways are **INTERCONNECTED** and **RECIPROCAL**

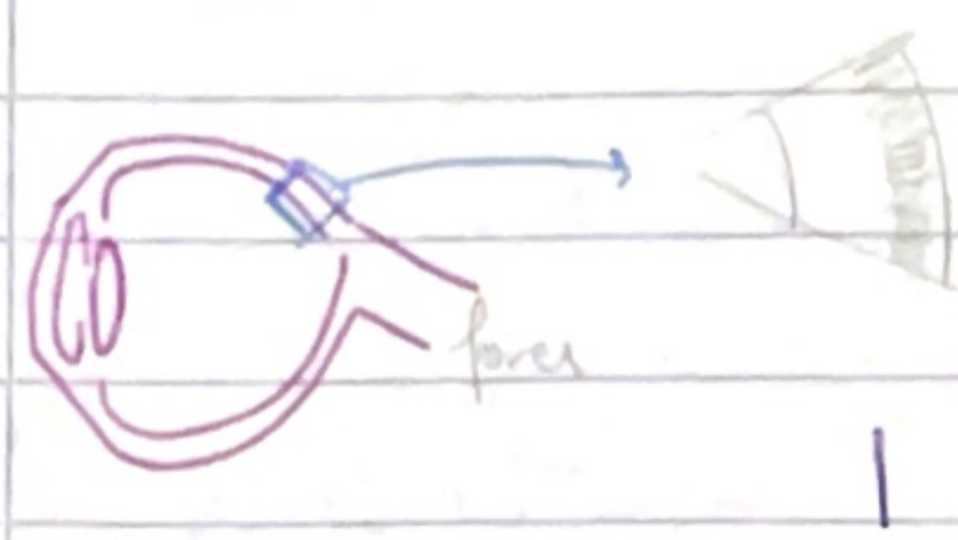
pulvinar area in the thalamus as relay

Receptive field:

Off area → fire when the ligh in the center turned off

On area → it fires when a spot of light is turned on there (~10°)

lateral inhibition



more near the fovea → 0,1° (smallest receptive field)

ECCENTRICITY

The ~~...~~ receptive field changes along the visual pathway

+ The visual cortex (V1) is more dedicated to the 10° central of visual space

+• Visual Cortex is organized into columns of specialized neurons.

↓

Visuotopic organization of the cells.

This neurons respond better to contrast